

LLM evaluation techniques

How to Measure, Compare, and Improve Large Language Models

Why LLM evaluation matters

LLM outputs are **probabilistic, not deterministic**

Same prompt → different answers

Traditional accuracy metrics don't always work

Why evaluation is critical

- Ensure **quality, correctness, and safety**
- Compare **models / prompts / versions**
- Reduce **hallucinations and bias**
- Validate **business impact**



LLM evaluation techniques can be grouped;

Automatic / Offline Metrics

LLM-as-a-Judge

Human Evaluation

Task-Based / Functional Evaluation

Safety, Bias & Robustness Evaluation

Online / Production Evaluation

LLM evaluation metrics

Correctness /
Accuracy

Relevance

Completeness

Faithfulness /
Groundedness

Hallucination
Rate

Semantic
Similarity

Fluency &
Readability

Coherence &
Logical Flow

Instruction
Following

Reasoning
Quality

Safety &
Toxicity

Correctness / Accuracy

- Whether the model's answer is **factually correct** or matches expected output.

How it's measured

- Exact Match (EM)
- F1 Score
- Comparison with ground truth

Best used for

- Q&A systems
- Fact-based tasks
- Structured outputs

Example

Q: What is the capital of France?

✓ Correct: Paris

✗ Incorrect: Lyon

Relevance

- Whether the response **directly answers the user's question**

Key focus

- No unnecessary information
- No missing intent

Best used for

- Chatbots
- Search & RAG systems

Example

User asks: *"Explain overfitting"*

- ✗ Model explains underfitting
- ✓ Model explains overfitting with examples

Completeness

- Whether the response covers **all important aspects** of the question

Example

Question: *“Explain Bias–Variance Tradeoff”*

- ✓ Explains bias
- ✓ Explains variance
- ✓ Explains tradeoff relationship

Best used for

- Educational content
- Documentation generation
- Analyst copilots

Faithfulness / Groundedness

- Whether the answer is **supported by provided context or source documents**

Critical for

- RAG

Example

- Answer only uses retrieved documents
- No external or fabricated facts

Best used for

- Groundedness score
- Context overlap

Hallucination Rate

- Frequency of **incorrect or fabricated information**

Types

- Fabricated facts
- Wrong citations
- Confident but false statements

Metric

- % of unsupported claims

Why important

Hallucinations destroy trust in enterprise GenAI systems

Semantic Similarity

- Meaning similarity between generated output and reference answer

Techniques

- Embedding cosine similarity
- BERTScore

Example

- “Car” vs “Automobile” → High similarity

Best used for

- Summarization
- Paraphrasing
- Rewriting tasks

Fluency & Readability

- It measures grammar, sentence flow, and natural language quality

Key checks

- Grammatically correct
- Easy to read
- Well-structured

Example

- ✗ Broken sentences
- ✓ Clear, professional language



Coherence & Logical Flow

- Whether the response follows a **logical progression**

Example

- Step-by-step reasoning
- No contradictions

Best used for

- Explanations
- Multi-step reasoning tasks

Instruction Following

- Whether the model follows **explicit user instructions**

Examples

- “Answer in bullet points”
- “Limit to 3 lines”
- “Respond in JSON format”

Failure example

- User asks for JSON → model returns plain text



Reasoning Quality

it measures ability to perform **logical, mathematical, or causal reasoning**

Evaluation aspects

- Step correctness
- Logical consistency
- Intermediate reasoning validity

Used in

- Math problems
- Decision support systems
- Analytics copilots



Safety & Toxicity

it measures harmful, biased, or unsafe outputs

Includes

- Hate speech
- Offensive content
- PII leakage

Critical for

- Public-facing applications
- Enterprise compliance

Thank-you!

Let's build LLM-as-a-judge!

